

Yakshith Daggupati

Hyderabad, India • daggupatiyakshithnaidu@gmail.com
github.com/Yakshith15 • linkedin.com/in/yakshithnaidu • yakshith15.github.io

EXPERIENCE

JPMorgan Chase

Jul 2025 – Present

Software Engineer I

Hyderabad, India

API Marketplace Platform. Teams publish APIs as Envoy-fronted proxies and other teams subscribe under quota, across ~4,000 APIs, ~40,000 proxies, and ~100,000 subscriptions. My team owns the system of record for every API, its analytics, and search.

- Built catalog search across ~4,000 production APIs, combining lexical (BM25) and semantic vector retrieval behind a per-query router. Relevance rose ~40% over the lexical-only baseline.
- Built the pipeline joining design-time data (API metadata, versions, proxies, subscriptions) with runtime telemetry from Envoy logs, attributing every request to an API, version, and subscriber. The platform's analytics dashboards run on it.
- Integrated gRPC into the Go service that enforces per-subscription quotas, replacing its REST interface: p99 latency down ~60%, throughput roughly doubled.
- Work across the platform stack: portal frontend, backend services, and the infrastructure they run on.

Software Engineer Intern

Jan 2025 – Jul 2025

- Migrated critical secrets from Kubernetes Secrets, which are only base64-encoded, to AWS Secrets Manager for encryption at rest, IAM-scoped access, and audit logging.
- Built frontend features for the marketplace portal, the surface teams use to publish APIs and manage subscriptions.

PROJECTS

Google File System, built from scratch

Go

- Reimplementing the GFS paper end to end: a master coordinating chunk metadata and leases, chunkservers storing 64 MB chunks, and a client library handling lookup and streaming reads and writes.
- Exercises recovery paths through simulated chunkserver crashes, and uses lease-based primary writes to hold the paper's consistency guarantees.

MIT 6.824 Distributed Systems Labs

Go

- Implemented MapReduce with task reassignment on worker failure, and Raft from the paper: leader election, log replication under partition, persistence, and snapshots.

EDUCATION

Chaitanya Bharathi Institute of Technology

Dec 2021 – May 2025

B.E., Computer Science Engineering (AI & ML) • 9.4 / 10 CGPA

TECHNICAL SKILLS

Languages: Go, Java, Python, TypeScript, JavaScript, C++

Infrastructure: AWS, Docker, Kubernetes, gRPC, CI/CD

Data: PostgreSQL, Redis, ETL pipelines, vector and lexical search

Focus: Distributed systems, backend services, system design